

## Module (04) :- The Transition From Analog to Digital Txn.

### Introduction :-

Communication system is of 2 types

- Analog comm<sup>n</sup> system [ACS]
- Digital communication system [DCS]

→ The purpose of communication system is to transfer the data from one place to another place.

→ ACS Transmits analog sig

Eg:- voice

→ In DCS, digital signals are transmitted

Eg:- Computer o/p, Optical drives

→ Eg for Digital comm<sup>n</sup> system are mobile comm<sup>n</sup> s/m, Satellite comm<sup>n</sup> s/m, Micro wave comm<sup>n</sup> s/m. etc

→ Digital sig are obtained by processing analog sig.

### How to Convert Analog sig to Digital sig

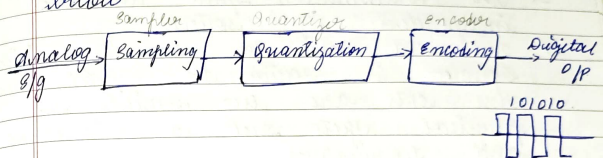
Any given analog sig can be converted into digital sig by combining the following three different basic operations such as

- Sampling
- Quantization

### iii) Encoding

→ The device which performs this operation is called as Analog to Digital Converter ADC.

→ The block diagram of ADC is shown below



### i) Sampling

The process of converting the continuous time signals into discrete time signals.

→ Sampling process results in sequence of analog sampled values.

### ii) Quantization

It is a process of converting sequence of analog sample values into a finite set of discrete values.

### iii) Encoding

It is a process of converting quantized sig values into a specific digital format.

Why DCS is more popular than ACS / Advantages of DCS over ACS

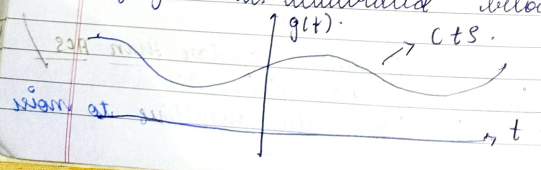
→ In DCS are less sensitive to noise



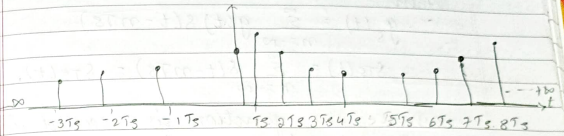
- and interference
- with digital system it is easier to integrate different services.  
eg: we can combine video & sound track into same channel
- The transmission speed can be independent of source
- The ckt used for handling digital signals is easier to repeat and these ckt are less sensitive to physical effects such as vibrations and temperature
- Digital sig are very simple to characterise and very easy to design.
- In digital system it is very easy to implement, the technique called as multiplexing.
- In digital system it is possible to control redundancy.
- In digital system the error correction and detection is possible
- RS provides high security to transmitted signal using techniques called as encryption and decryption
- RS provides improved signal quality

### Sampling Process

but we consider arbitrary sig  $g(t)$  as illustrated below.



suppose that we sampled  $g(t)$  instantaneously at a uniform rate, once every  $T_s$  seconds, as illustrated below:



The sampling process results infinite sequence of samples spaced  $T_s$  seconds apart and represented by  $\{g(nT_s)\}$ , where

$$n = 0, \pm 1, \pm 2, \dots, \pm \infty$$

$T_s \rightarrow$  sampling interval.

The reciprocal of sampling period is called sampling rate represented by  $f_s = 1/T_s$

This ideal form of sampling is called as instantaneous sampling. Let  $g_s(t)$  represents the signal obtained by the sampling process and is represented in terms of direct delta just as follows

$$g_s(t) = \sum_{n=-\infty}^{\infty} g(nT_s) \delta(t - nT_s) \quad \text{--- (1)}$$

$g_s(t) \rightarrow$  ideal sampled sig.  
 $\delta(t - nT_s) =$  Delta func occurring at time  $(t - nT_s) = 0$ .  
(or)  
 $t = nT_s$ .

where,

$$g(nT_s) = \text{sampler value of } g(t) \text{ at } t = nT_s$$



hence we can rewrite eqn ① as follows:

$$\therefore g(t) = g(t_s)$$

then

$$g_s(t) = \sum_{n=-\infty}^{\infty} g(t) \delta(t - nT_s)$$

$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) = s(t_s)$$

where the function is called ideal sampling fun.  
Hence,

the above eqn can be rewritten in terms of ideal sampling fun<sup>c</sup> as follows -

$$g_s(t) = s(t_s) \cdot g(t) \quad \text{--- ②}$$

Taking Fourier Transform on BS, we get

$$G_s(f) = G(f) * f_s \sum_{m=-\infty}^{\infty} \delta(f - m f_s)$$

$$G_s(f) = f_s \sum_{m=-\infty}^{\infty} G(f) * \delta(f - m f_s)$$

using shifting property of Fourier Transform.  
WKT

$$G(f) * \delta(f - m f_s) = G(f - m f_s)$$

$$\therefore G_s(f) = f_s \sum_{m=-\infty}^{\infty} G(f - m f_s) \quad \text{--- ③}$$

$G_s(f) \rightarrow$  spectrum of sampled s/g

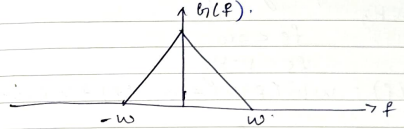
$f_s \rightarrow$  sampling Rate =  $1/T_s$ .

$G(f) \rightarrow$  spectrum of a I/P given signal

then, Hence eq ③ represents the process of uniformly sampling a continuously time signal, that results a periodic spectrum with a period = sampled rate ( $f_s$ ).

Suppose the s/g  $g(t)$  is strictly band limited s/g with no frequency components higher than  $w$  Hz.

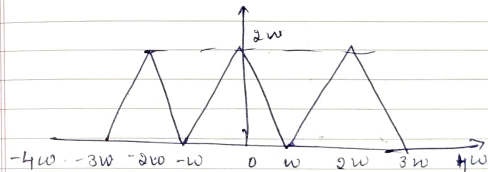
As illustrated below:



But we study different 3 cases:

case(i) :-

The spectrum of a sampled s/g  $G_s(f)$  for a sampling rate  $f_s = 2w$  Hz is illustrated below.



This case is called Nyquist sampling / (or) exact sampling

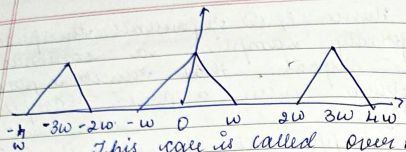
case(ii) :-

Spectrum of a sampled s/g  $G_s(f)$  for a sampling rate  $f_s$  greater than  $2w$  Hz.  $f_s > 2w$

$f_s = 3w$  Hz.

$$G_s(f) = 3w G(f + 3w) + 3w G(f) + 3w G(f - 3w)$$





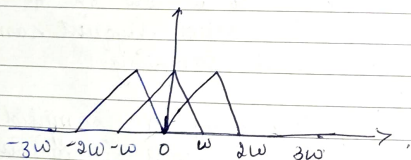
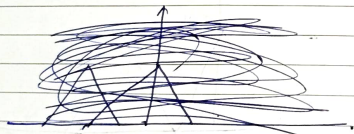
This case is called oversampling.

ex (iii):- spectrum of a sampled s/g for the sampling rate  $f_s$  less than  $2W\text{Hz}$

$$f_s < 2W$$

$$f_s = W\text{Hz}$$

$$G_s(f) = W G(f+W) + W G(f) + W G(f-W)$$



aliasing (or) distortion

This case is called undersampling.  
(WSE-3) Because of distortion, it is not possible to recover  $m(t)$  at receiver's s/g

side if it is undersampling.  
But it can be recovered in case of negligent and over sampling.

A band limited s/g of finite energy which has frequency components less than  $W\text{Hz}$  is completely removed from a knowledge of its samples taken at a rate of  $f_s = 2W$  samples per second.

Let us consider another useful expression for Fourier transform of  $G_s(f)$ . This expression may be obtained by taking Fourier transform of eqn (i)

Taking Fourier transform of a above eqn we get

$$g_s(t) = \sum_{n=-\infty}^{\infty} g(nT_s) \cdot \delta(t - nT_s)$$

$$G_s(f) = \sum_{n=-\infty}^{\infty} g(nT_s) e^{j2\pi nT_s f}$$

Consider the sampling period  $T_s = 1/2W$ , substituting this in the above eqn

$$G_s(f) = \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2W}\right) e^{j2\pi \frac{n}{2W} f} \quad \text{--- (4)}$$

W.K.T

$$G_s(f) = f_s \sum_{m=-\infty}^{\infty} G(f - m f_s) \quad \text{--- (5)}$$

for  $m=0$

$$G_s(f) = f_s G(f) + f_s \sum_{\substack{m=-\infty \\ m \neq 0}}^{\infty} G(f - m f_s) \quad \text{--- (6)}$$

Hence under the following two cond<sup>n</sup>



$$b(f) = 0 \quad ; \quad |f| > w \text{ Hz}$$

②  $f_s = 2w \text{ Hz}$  is reduced to  
 $\therefore$  eq<sup>n</sup> (b)  $b_s(f) = f_s b(f) \quad -w \leq f \leq w$   
 Modify this eq<sup>n</sup> to get eq<sup>n</sup> for  $b(f)$   
 $b(f) = \frac{1}{f_s} b_s(f)$

Substitute (u) in the above eq<sup>n</sup>.

$$b(f) = \frac{1}{2w} \left\{ \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2w}\right) e^{-j\pi n f} \right\} \quad ; \quad -w \leq f \leq w \quad \text{--- (3)}$$

To reconstruct the s/g  $g(t)$  we need to take inverse Fourier transform of  $b(f)$  i.e.,  
 $g(t) = \text{IFT}[b(f)]$

$$= \int_{-w}^w b(f) e^{j2\pi f t} df$$

$$g(t) = \int_{-w}^w \left[ \frac{1}{2w} \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2w}\right) e^{-j\pi n f} \right] e^{j2\pi f t} df$$

Interchanging the order of integration and summations we get

$$g(t) = \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2w}\right) \cdot \frac{1}{2w} \int_{-w}^w e^{j2\pi \left(t - \frac{n}{2w}\right) f} df$$

$$g(t) = \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2w}\right) \cdot \frac{1}{2w} \left[ \frac{e^{j2\pi \left(t - \frac{n}{2w}\right) f}}{j2\pi \left(t - \frac{n}{2w}\right)} \right]_{-w}^w$$

$$\text{--- (4)}$$

$$= \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2w}\right) \cdot \frac{1}{2w} \left\{ \frac{e^{j2\pi \left(t - \frac{n}{2w}\right) w} - e^{-j2\pi \left(t - \frac{n}{2w}\right) w}}{j2\pi \left(t - \frac{n}{2w}\right)} \right\}$$

$$= \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2w}\right) \cdot \frac{1}{2w} \left\{ \frac{e^{j\pi \left(2wt - n\right)} - e^{-j\pi \left(2wt - n\right)}}{j2\pi \left(2wt - n\right)} \right\}$$

$$= \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2w}\right) \cdot \frac{1}{\pi(2wt - n)} \frac{e^{j\pi(2wt - n)} - e^{-j\pi(2wt - n)}}{2j}$$

$$\text{W.K.T.} \quad \frac{e^{j\theta} - e^{-j\theta}}{2j} = \sin \theta$$

$$= \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2w}\right) \cdot \frac{1}{\pi(2wt - n)} \sin[\pi(2wt - n)]$$

can be simplified further using sinc func<sup>n</sup> given by  
 $\text{sinc}(x) = \frac{\sin \pi x}{\pi x}$

$$= \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2w}\right) \frac{\sin[\pi(2wt - n)]}{\pi(2wt - n)}$$

This resulting eq<sup>n</sup> is called as a interpolation eq<sup>n</sup>. This eq<sup>n</sup> is used to reconstruct the s/g from the analog sample values.

Statement of Sampling Theorem: -

A bandlimited s/g of a finite energy which has no frequency components higher than  $w \text{ Hz}$  is completely described by its analog sampled values defined at a rate  $f_s \geq 2w \text{ Hz}$   
 (or)  $T_s = 1/2w \text{ s apart}$



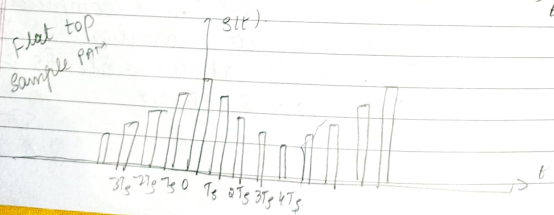
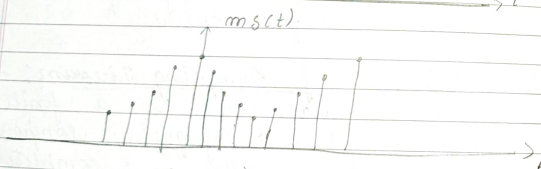
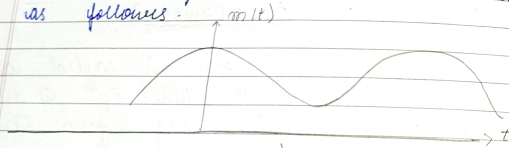
## Pulse Amplitude Modulation (PAM)

In PAM the amplitude of pulses are varied in proportional to the corresponding sample values of  $m(t)$  i.e., continuous time s/g.

Pulses can be of rectangular form. The PAM waveform can be generated by combining the following two operations.

- i) Instantaneous sampling of message  $s/g$  at every  $T_s$  second, where the sampling rate  $f_s = 1/T_s$ .
- ii) Lengthening the duration of each analog sample value so as to obtain a pulse of duration  $T$  second where  $T < T_s$ .

These operations are illustrated as follows.



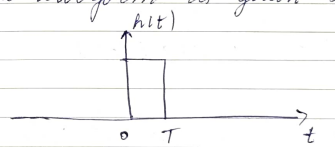
Let  $s(t)$  denotes the sequence of flat top pulses generated in the manner described above. Hence we may represent eq<sup>n</sup> of PAM wave as follows

$$s(t) = \sum_{n=-\infty}^{\infty} m(nT_s) * h(t - nT_s) \quad \text{--- (1)}$$

$T_s$  = sampling interval  
 $m(nT_s)$  = analog sample values of  $m(t)$  at time  $t = nT_s$   
 $h(t)$  = rectangular pulse of duration  $T$  and it is defined as follows

$$h(t) = \begin{cases} 1 & 0 < t < T \\ \frac{1}{2} & t = 0 \text{ \& } t = T \\ 0 & \text{elsewhere} \end{cases}$$

The waveform is given by.



WKT instantaneous sampled version of  $m(t)$  is given by  $ms(t) = \sum_{n=-\infty}^{\infty} m(nT_s) * \delta(t - nT_s)$  --- (2)

Let us consider the instant convolution of  $ms(t)$  and  $h(t)$

$$ms(t) * h(t) = \int_{-\infty}^{\infty} ms(\tau) * h(t - \tau) d\tau$$

Substitute eq<sup>n</sup> (2)



$$m(t) * h(t) = \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} m(nTs) \delta(t-nTs) \int_{-\infty}^{\infty} h(t-\tau) d\tau$$

After interchanging the order of integration & summation we can rewrite the eq<sup>n</sup> as follows

$$= \sum_{n=-\infty}^{\infty} m(nTs) \int_{-\infty}^{\infty} \delta(t-nTs) \cdot h(t-\tau) d\tau$$

The last part of the resulting eq<sup>n</sup> can be simplified using shifting property of  $\delta$  fun<sup>y</sup>

$$= \int_{-\infty}^{\infty} \delta(t-nTs) h(t-\tau) d\tau = h(t-nTs)$$

$$m(t) * h(t) = \sum_{n=-\infty}^{\infty} m(nTs) h(t-nTs) \quad \text{--- (3)}$$

Combining eq<sup>n</sup> (1) & (3)

$$s(t) = m(t) * h(t)$$

The above eq<sup>n</sup> indicates pulse-amplitude modulated sig<sup>n</sup>  $s(t)$  is eq<sup>n</sup> to the convolution of  $m(t)$  &  $h(t)$ .

Taking Fourier Transform of above eq<sup>n</sup> we get

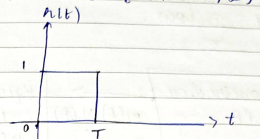
$$S(f) = M(f) \cdot H(f)$$

$$S(f) = f_s \sum_{m=-\infty}^{\infty} M(f - mf_s) \cdot H(f)$$

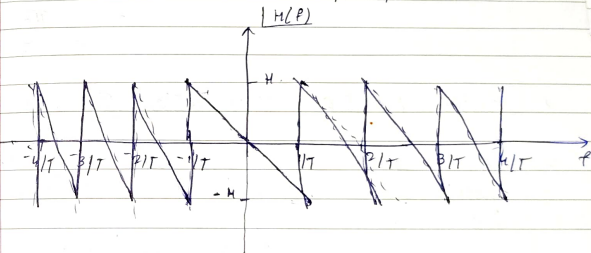
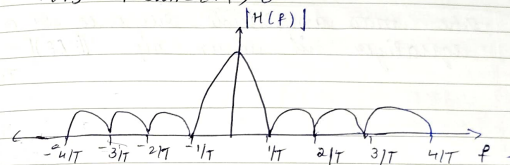
where  $FT[m(t)] = f_s \sum_{m=-\infty}^{\infty} M(f - mf_s)$

$$FT[h(t)] = H(f)$$

$$h(t) = \text{rect}(t/T - 1/2)$$



$$H(f) = T \text{sinc}(Tf) e^{-j\pi f T}$$

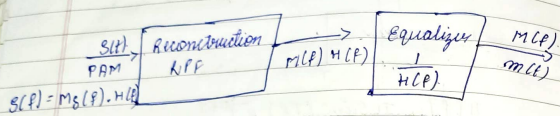


In pulse amplitude modulation technique we introduced amplitude distortion as well as delay of  $T/2$  seconds. This distortion is caused by lengthening the duration of each sample. This distortion is called aperture effect.

This distortion can be corrected by connecting equalizer in cascade



with reconstruction low pass filter at the receiver side and it is illustrated below.



The mag amplitude response of a equalizer is given by  $|H(f)| = \frac{1}{T \text{sinc}(fT)}$

$$= \frac{1}{T \frac{\sin(\pi f T)}{\pi f T}}$$

$$= \frac{\pi f T}{\sin(\pi f T)}$$

### \* Pulse Position Modulation Technique:-

→ In PPM the position of the pulse is varied in accordance with the message s/g.

In PPM wave generation, first PWM is obtained by using the message signal  $m(t)$  and pulse carrier  $g(t)$ . This process is illustrated below.

Let  $T_s$  denote the sampling duration. Using the sample  $m(nT_s)$  of  $m(t)$  to modulate position of  $n^{th}$  pulse we obtain the PPM signal.

$$S(t)_{PPM} = \sum_{n=-\infty}^{\infty} g(t - nT_s - K_p m(nT_s))$$

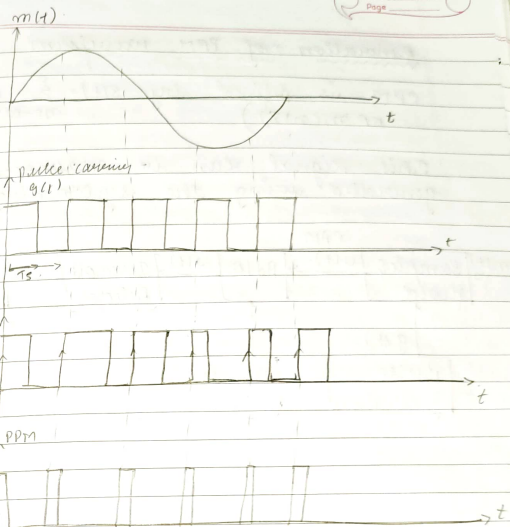
where  $K_p$  = sensitivity of PPM.

$g(t)$  = pulse.

→ For non-overlapping PPM it is necessary to satisfy the following condition i.e.,  $g(t) = 0$  for  $t > T_s/2$ .

i)  $|t| > \frac{T_s}{2} - K_p |m(t)|$  and  $K_p |m(t)| < \frac{T_s}{2}$

ii)  $K_p |m(t)|_{max}$

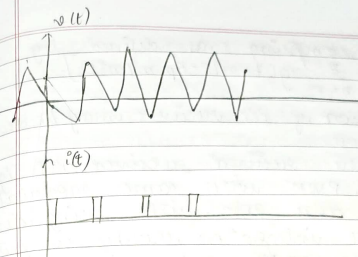
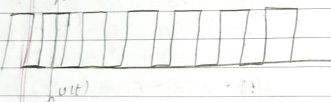
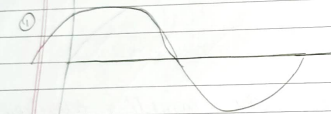
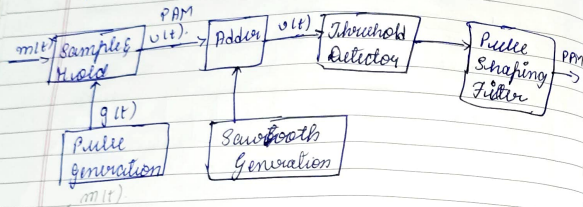




### Generation of PPM waveform

PPM is defined by  $s(t) = \sum_{n=-\infty}^{\infty} g(t - nT_s - KPm(nT_s))$

This signal can be regenerated using the system shown below



Step (1): The message sig  $m(t)$  is converted into PAM signal by means of sample and hold circuit by generating a staircase waveform  $v(t)$ . This operation is illustrated in fig 3.

Step (2): The signal  $v(t)$  and sawtooth waveform are added, the resulting combined sig  $v(t)$  is shown in fig 4.

Step (3): The combined sig  $v(t)$  is applied to the threshold detector that produces a very narrow samples when each time  $v(t)$  crosses the level in the going direction as shown in the fig 5.

Step (4): Finally PPM sig  $s(t)$  is generated by using the sequence of narrow pulses through a pulse shaping filter.

### Demodulation / Detection of PPM waves.

But we consider a PPM wave  $s(t)$  with

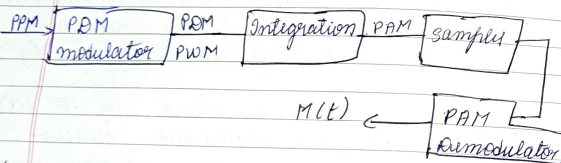


uniform sampling is defined by  

$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s - k_p m(nT_s))$$

The operation of PPM receiver may be divided as follows.

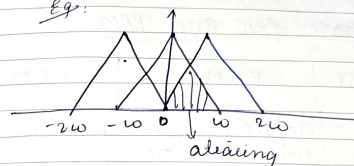
1. Convert the received received PPM s/g into PDM. PWM with same modulator
2. Integrate this PDM with using divider of finite integration thereby computing area under each pulse of the PDM.
3. The sampled o/p of integrator at a uniform rate to produce PAM s/g waveform
4. Then, demodulate the PAM wave to recover the message s/g. This operation is illustrated below.



Explain aliasing and describe the method to avoid it.

Aliasing is a kind of distortion that occurs during the sampling process when the sampling rate is less than  $2W$  Hz.

Eg:



Note: Under this condition (under sampling) it is not possible to recover the  $m(t)$  at the receiver's side.

This distortion can be completely eliminated or avoided by sampling rate  $f_s \geq 2W$  Hz, so that we can recover the message s/g at receiver side without any problem.

Q. Describe the advantages and disadvantages of Digital communication system

⇒ Disadvantages:

- 1) System complexity is more
- 2) Requires larger bandwidth
- 3) It results in Quantization Errors
- 4) Synchronization issues occur
- 5) More power consumption.

|                               | Natural Sampling                                                        | Flat Top Sampling                                                          |
|-------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Definition                    | Sampling in which the top of the pulse follows the shape of the I/P s/g | Sampling in which the sampled value is held constant during pulse duration |
| Pulse Shape                   | Non-Flat top                                                            | Flat top                                                                   |
| Signal variation during pulse | varies continuously                                                     | Remains constant                                                           |



|              |                                    |                                      |
|--------------|------------------------------------|--------------------------------------|
| Method       | sig is multiplied with pulse train | uses sample & hold ckt               |
| Complexity   | more difficult to implement        | easier to implement                  |
| Applications | Communications & many studies      | PCM and Digital Communication System |

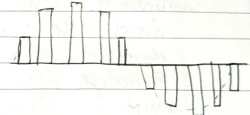
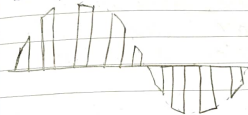
### \* Compare PPM and PAM

| => Features           | PAM                                          | PPM                                                                            |
|-----------------------|----------------------------------------------|--------------------------------------------------------------------------------|
| Explanation           | Amplitude of pulse varies a/c to message sig | Position of pulse varies a/c to message sig                                    |
| Power Requirement     | Higher power is required                     | Lower power is required                                                        |
| Bandwidth Requirement | Lower bandwidth is required                  | Higher bandwidth is required                                                   |
| Noise Immunity        | Less immune to noise                         | Better noise immunity                                                          |
| Efficiency            | Less power efficient                         | More power efficient                                                           |
| Applications          | PCM systems, telemetry, radio comm systems   | Optical communication, telemetry, Advanced comm systems like optical fibre etc |
| Complexity            | System complexity is less                    | More complex                                                                   |

### Comparison b/w Natural & Flat top sampling

| Natural Sampling                                      | Flat top sampling                                                                       |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------|
| The amplitude of sampled sig follows the message sig. | → In this, the amplitude of sampled sig is held constant for a duration of $T$ seconds. |

#### Waveform:-



|                                                         |                                                            |
|---------------------------------------------------------|------------------------------------------------------------|
| → It is more accurate                                   | → Less in accuracy                                         |
| → More complex                                          | → Less complex                                             |
| → Used analog switching ckt to realize natural sampling | → Sample and hold ckt is used to realize flat top sampling |
| → Rarely used                                           | → Widely used                                              |

### Bandwidth - Noise Trade off:-

In the context of noise performance compared to all the pulse modulation techniques such as PAM, PPM, PWM. A PPM system represents the optimum form of analog pulse modulation system.

In CWM ~~FM~~ FM system exhibits

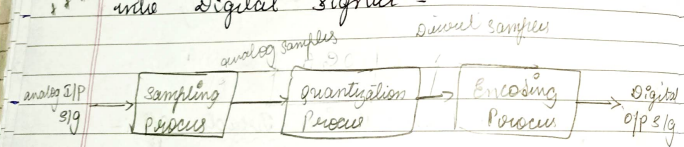


a better noise performance compared to AM and FM systems.

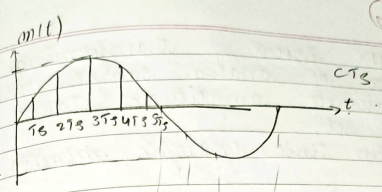
- The PPM system exhibits a similar noise performance of FM system.
- Both PPM and FM systems have figure of merit proportional to the square of the bandwidth normalized w.r.t message signal bandwidth.
- In both the systems in order to improve the noise performance we need to increase the transmission bandwidth. This trade off can be used in CWM and analog pulse modulation systems.

### Quantization Process:-

Q) With the neat diagram explain the conversion of analog sig into Digital signal.



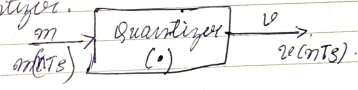
but we consider continuous time sig  $m(t)$  having finite amplitude range from  $3.5V$  to  $-3.5V$ , as illustrated below



but we assume that this sig is sampled at uniform rate  $f_s = 1/T_s$  Hz. This operation results sequence of analog sample values, as illustrated below.



say  $\{m(nT_s)\} = \{0, 1.1, 2.2, 3.3, 2.3, 0.9, \dots\}$  these analog sample values are now provided as input to the quantizer.



In quantization process the message sig  $m(t)$  is divided into a  $L$  equal amplitude levels to obtain the finite set. In this example we have considered  $L=8$ . Hence the finite set of discrete amplitude levels is defined as

$$= \{3.5, 2.5, 1.5, 0.5, -0.5, -1.5, -2.5, -3.5\}$$

Amplitude quantization is defined



as the process of transforming the sequence of analog samples into a finite set of discrete amplitudes taken from a finite set.

In this example  $m(t)$

| Quantizer i/p<br>$m(nT_s)$ or $m$ | Quantizer o/p<br>$\hat{v}(nT_s)$ or $\hat{v}$ |
|-----------------------------------|-----------------------------------------------|
| 0                                 | 0.5                                           |
| 1.1                               | 1.5                                           |
| 2.2                               | 2.5                                           |
| 3.3                               | 3.5                                           |
| 2.3                               | 2.5                                           |
| 0.9                               | 0.5                                           |

→ The quantizer inputs are called division levels represented by  $M_k$  where  $k = 1, 2, 3, \dots, K$  where  $K$  represents number of discrete amplitude levels.

→ Quantizer o/p are called as representation / reconstruction levels.

→ Step size is the spacing b/w two two adjacent representation levels is called as step size and represented by  $\Delta$ . In this example  $\Delta$  size is 1. In this example  $\Delta$  is computed by using the eq<sup>n</sup>  $\Delta = \frac{2M_{max}}{K}$

→ In quantiz<sup>n</sup> process the analog samples value  $m$  are mapped to discrete amplitude values  $\hat{v}$ . This mapping is defined as follows

$$\hat{v} = q(m)$$

→ It is the quantizer characteristics and is a staircase function by definition.

### Different Types of Quantizer:-

2 types of Quantizer  
① uniform quantizer.  
② Non uniform quantizer

→ In uniform quantizer the step size is constant (or) fixed

\* Non uniform quantizer the step size  $\Delta k$  is not constant, it varies with  $m(t)$

### Different Types of Uniform Quantizer

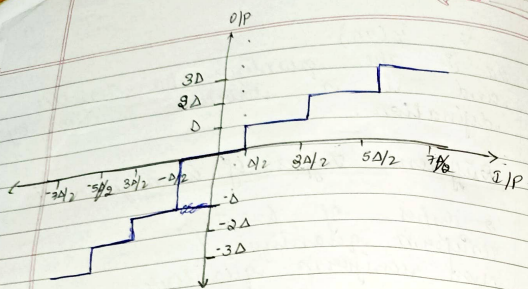
There are two different types of a uniform quantizer.

i) Mid Tread type of uniform quantizer  
ii) Mid Rise " " " " " "

### i) Mid Tread type of uniform quantizer:-

Input-output characteristics of uniform quantizer of mid Tread type is shown below.

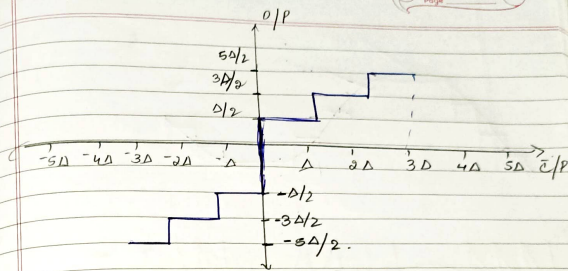




- In this quantizer the decision levels (i/p) are located at  $\pm \Delta/2, \pm 3\Delta/2, \pm 5\Delta/2, \dots$  and representation levels (o/p.s) are located at  $0, \pm \Delta, \pm 2\Delta, \pm 3\Delta, \dots$
- If i/p lies in the range  $-\Delta/2$  to  $\Delta/2$  then it is marked to 0.
  - If i/p is in the range  $\Delta/2$  to  $3\Delta/2$  then it is marked to  $\Delta$ .
  - From this characteristic curve we can notice that origin lies in middle of tread of the staircase graph. Hence this quantizer is called as mid-rise type uniform quantizer.

## ii) Mid-rise type of uniform quantizer:-

I/P O/P characteristics of mid rise type of uniform quantizer is shown below.



- In this quantizer the decision levels (i/p) are located at  $\pm \Delta, \pm 2\Delta, \pm 3\Delta, \dots$  and representation levels (o/p) are located at  $0, \pm \Delta/2, \pm 3\Delta/2, \pm 5\Delta/2, \dots$
- If i/p is 0 to  $\Delta$  then it is mapped to  $\Delta/2$ , if the i/p is  $0$  to  $-\Delta$  then it is mapped to  $-\Delta/2$  and so on.
- From the characteristic curve we can notice that the origin lies in the middle of staircase graph. Hence this quantizer is called as mid-rise type of uniform quantizer.

## Quantization Noise:-

The use of quantization introduces error defined as difference b/w quantizer i/p [m] and quantizer o/p [V]. This error is called as quantization noise (or) error.